EXECUTIVE PRESENTATION · SAMPLE WORKBOOK

From weekly sales data to reviewable price decisions

A local, reproducible pipeline links historical KPIs, a 26-week sales forecast, and guarded next-week price proposals.

30
products

260
observed weeks

11
proposed price increases

> Decision today: review the proposals and pilot measurement plan before any live price change.

01 · DATA PIPELINE

Traceable by design, from Excel to decisions

Any compatible workbook is archived byte-for-byte; every parsed row is retained before cleaning.

INPUT
Excel
Recognized headers across worksheets; no fixed product or week count.

→

PERMANENT RECORD
Raw SQLite
Workbook BLOB, original rows, source locations, quality issues.

→

VALIDATED
Star schema
Product & week dimensions; cleaned weekly fact with provenance.

→

ANALYSIS
Versioned outputs
KPI views, backtests, forecasts, price recommendation runs.

→

USE
Dashboard
Local Streamlit readout and CSV export for review.

7,839
raw rows archived

7,800
clean product-weeks

39
equivalent duplicates

16
sales values recovered

Invalid required values are rejected from the clean fact but remain in raw history. The supplied sample has **zero rejected rows**.

Source: README.md · sql/schema.sql

02 · FORECASTING APPROACH

Rolling validation selects a random forest

Direct 26-week sales forecasts; product history and signals known at each forecast origin only.

Development MAE · units per product-week ↓



Across 3 selection windows · 2,340 product-weeks. Chosen by pooled MAE, 5.4% below the best boosting candidate.

SELECTED MODEL

47.51%

Development WAPE; RMSE 814.0 units

Compared with two simple baselines, five boosting losses and three forest split criteria.

26-week output

1,019,745 units

780 product-week predictions for relative weeks 261–286. Forecasts approximate recorded sales, not unconstrained demand.

> The fourth historical window was previously inspected. It is a retrospective diagnostic, not a fresh untouched holdout.

Source: reports/forecast_summary.md · reports/forecast_evaluation.json

03 · CORE KPI INSIGHTS

Recent revenue grew; inventory needs review

Weeks 248–260 versus the preceding complete 13-week period. Amounts use the workbook's unspecified currency.

16.04m
recent revenue

+3.09%
revenue versus prior 13 weeks

21.49%
gross profit margin

489,871
units sold

Period comparison

Prior 13 weeks

15.56m

Latest 13 weeks

16.04m

Recent gross profit: 3.45m; source-reported operating profit: 1.88m.

LATEST OBSERVED WEEK · 260

−15.43% revenue WoW

8 products report zero opening stock; 9 sold more than their opening stock. In-week receipts are unknown, so these are diagnostics, not confirmed stockouts.

Margin uses **total gross profit ÷ total revenue**. Formal DSI, DSO and customer default risk need fields absent from this workbook.

Source: reports/kpi_summary.json · reports/kpi_definitions.md

04 · DASHBOARD WALKTHROUGH

One local view for the decision workflow

Review portfolio trends, drill into a product, inspect forecast and stock warnings, then export proposed prices.

Overview | Demand forecast | Inventory diagnostics | Pricing rules | Data and method

Filters

Product

All products ▾

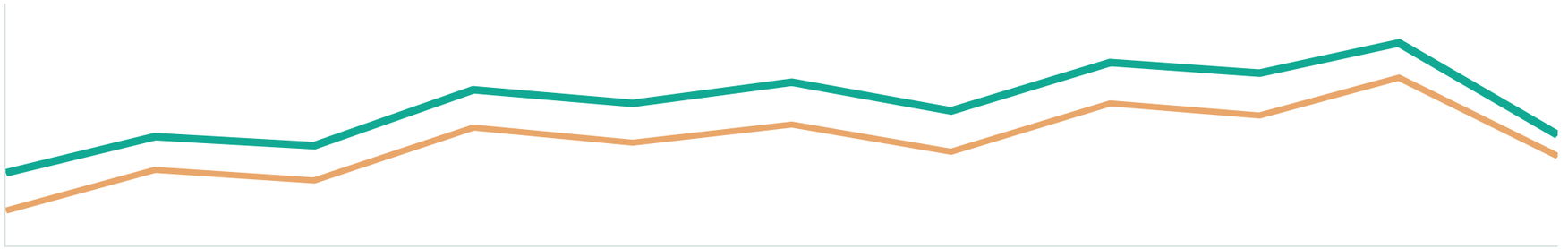
History window

13 weeks ▾

Switch product to compare its historical chart with future units.

OBSERVED WEEKS 248–260

16.04m	+3.09%	21.49%	489,871
Revenue	Prior period	Gross margin	Units



Illustrative layout of the local Streamlit tabs; chart line shapes are schematic.

Run locally after the README setup (PowerShell): `.\.venv\Scripts\python.exe -m streamlit run .\dashboard\app.py`. Page refresh does not retrain models.

Source: dashboard/app.py · README.md

05 · DYNAMIC PRICING LOGIC

Simple rules create auditable proposals

Next-week forecast versus the prior 13-week sales average triggers a small, bounded price move.

Forecast ≥ 1.20 × recent mean

Increase 3%

Subject to gross margin and move limits.

Forecast ≤ 0.80 × recent mean

Decrease 3%

Only where policy guardrails still hold.

Otherwise

Hold

Missing or zero reference sales normally hold; the margin floor may require an increase.

Hard guardrails

Minimum **15% gross unit margin** · at most **5% price change** per run · manual review when these constraints conflict.

Sample week 261 result

11 increases · 19 holds · 0 decreases · 0 manual reviews. **15 historical inventory warnings** prompt a stock check.

Break-even volume shows units needed to preserve reference gross profit at the proposed price. It is **not** a demand forecast at that price.

Source: reports/pricing_strategy.md · data/pricing_report.json

06 · PROJECTED BUSINESS ROI

Scenario economics depend on volume response

Illustrative one-week sensitivity for the 11 proposed increases; no price-response model or measured uplift is available.

Gross profit versus current-price reference

VOLUME AT NEW PRICES	Δ GROSS PROFIT	ROI*
0% fewer units	+12,579	+152%
5% fewer units	+7,286	+46%
10% fewer units	+1,993	-60%
15% fewer units	-3,300	-166%

* ROI = (incremental gross profit − assumed weekly cost) ÷ assumed weekly cost. Cost assumption: 5,000 in source currency units.

DECISION THRESHOLD

≈ 7.2% fewer units

At this volume loss, the scenario just covers the assumed 5,000 weekly cost. With no launch cost, gross profit breaks even at ≈ 11.9% fewer units.

What would make ROI real?

Approve actual policy limits and cost budget; verify current inventory; run a controlled price pilot; measure observed volume, gross profit and operating cost against a valid comparison group.

The ROI table is **conditional arithmetic**, not a validated forecast. Prices, costs and sales units are the latest available references; actual response and execution cost are unknown.

Source: data/pricing_report.json · reports/pricing_strategy.md · calculation: Σ[new units × (new price − unit cost)] − baseline gross profit